

Bitcoin Long-Term Valuation & Hybrid Forecasting

A Practical Data Analysis Portfolio Project

1. Project Overview

This project explores whether Bitcoin has a stable long-term price trend and whether we can use that trend to assess current valuation and make forward-looking estimates.

I built the analysis in three progressive stages:

A long-term power-law trend model

A valuation diagnostic layer using residual Z-scores

A hybrid model that attempts to adjust the long-term trend with short-term market factors

The goal was not to create a perfect trading signal, but to build a transparent, statistically grounded framework that can answer practical questions such as:

Is there a meaningful long-term structure in Bitcoin's price?

Where does the current price sit relative to that structure?

How useful are short-term factors in improving the forecast?

2. Motivation

Bitcoin's price history shows extreme volatility, yet many observers have noted that its long-term trajectory appears smoother when viewed on a logarithmic scale.

As someone with a background in applied statistics, I wanted to test this observation rigorously:

Can a simple statistical model capture a large portion of Bitcoin's long-term price movement?

Can we turn the model's residuals into a practical valuation indicator?

Does adding short-term volume and return factors meaningfully improve near-term forecasts?

These questions are relevant for research roles, investment analysis, and Web3 data positions where the ability to separate signal from noise matters more than producing overly complex models.

3. Data & Methodology

Data

Daily Bitcoin price data from genesis (2009) through early September 2026.

Stage 1 – Long-term Power Law Model

I modeled the relationship:

$$\log(\text{Price}) = \alpha + \beta \cdot \log(\text{Days since genesis})$$

This is equivalent to a power-law relationship of the form $\text{Price} \approx a \times \text{Days}^b$.

Ordinary Least Squares (OLS) was used for estimation.

Stage 2 – Valuation Z-Score

After fitting the power-law model, I calculated the residuals (actual log price minus predicted log price) and standardized them into a Z-score.

This Z-score serves as a simple valuation metric:

High positive values → price significantly above the long-term trend (potentially overheated)

High negative values → price significantly below the trend (potentially undervalued)

Stage 3 – Hybrid Short-term Adjustment

I attempted to predict changes in the Z-score using short-term factors (30-day volatility, log volume, lagged returns). The predicted residual path was then added back to the long-term trend to produce a hybrid price forecast for the remainder of 2026.

4. Key Results

Stage 1 – Power Law Model

Table 1. Power-Law Model Regression Results

Independent Variable	Coefficient (β)	Std. Error	t-statistic	p-value	95% Confidence Interval
Intercept (β_0)	-30.8601	0.311	-99.128	< 0.001	[-31.470, -30.250]
ln(Days) (β_1)	4.8423	0.037	131.114	< 0.001	[4.770, 4.915]

Table2. Power-Law Model Summary Statistics

Metric	Value
R-squared	0.830
Adjusted R-squared	0.829
Number of Observations	3,535
F-statistic	17,190
Prob (F-statistic)	< 0.001

The model explains approximately 83% of the variation in log price.
Estimated exponent ≈ 4.84 , indicating strong long-term growth.

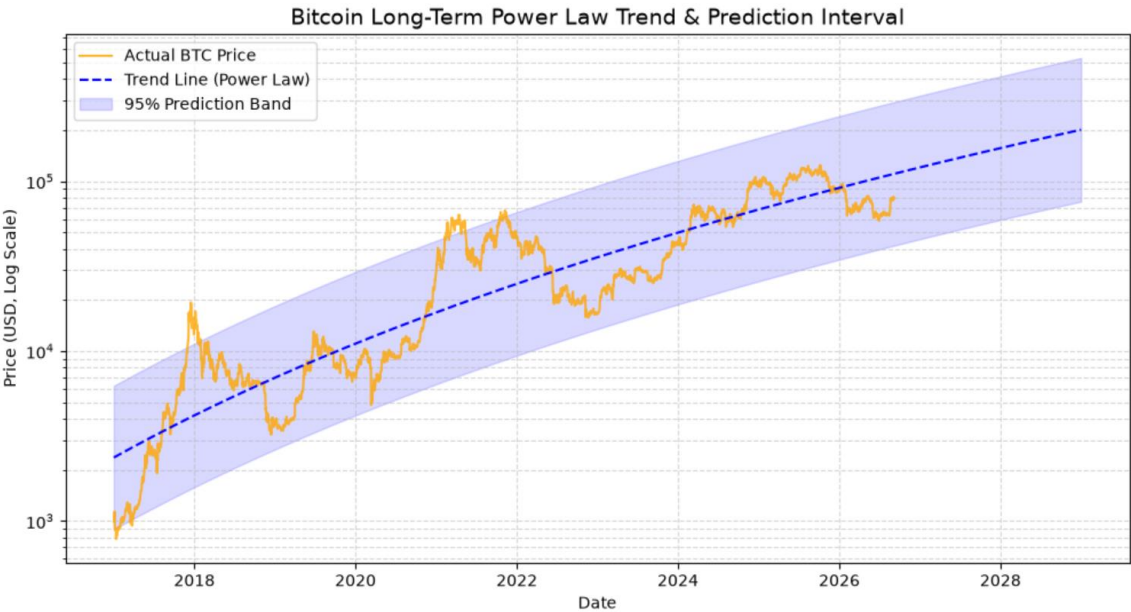


Figure 1. Bitcoin long-term power-law trend and 95% prediction band.

Stage 2 – Current Valuation (as of 5 Sep 2026)

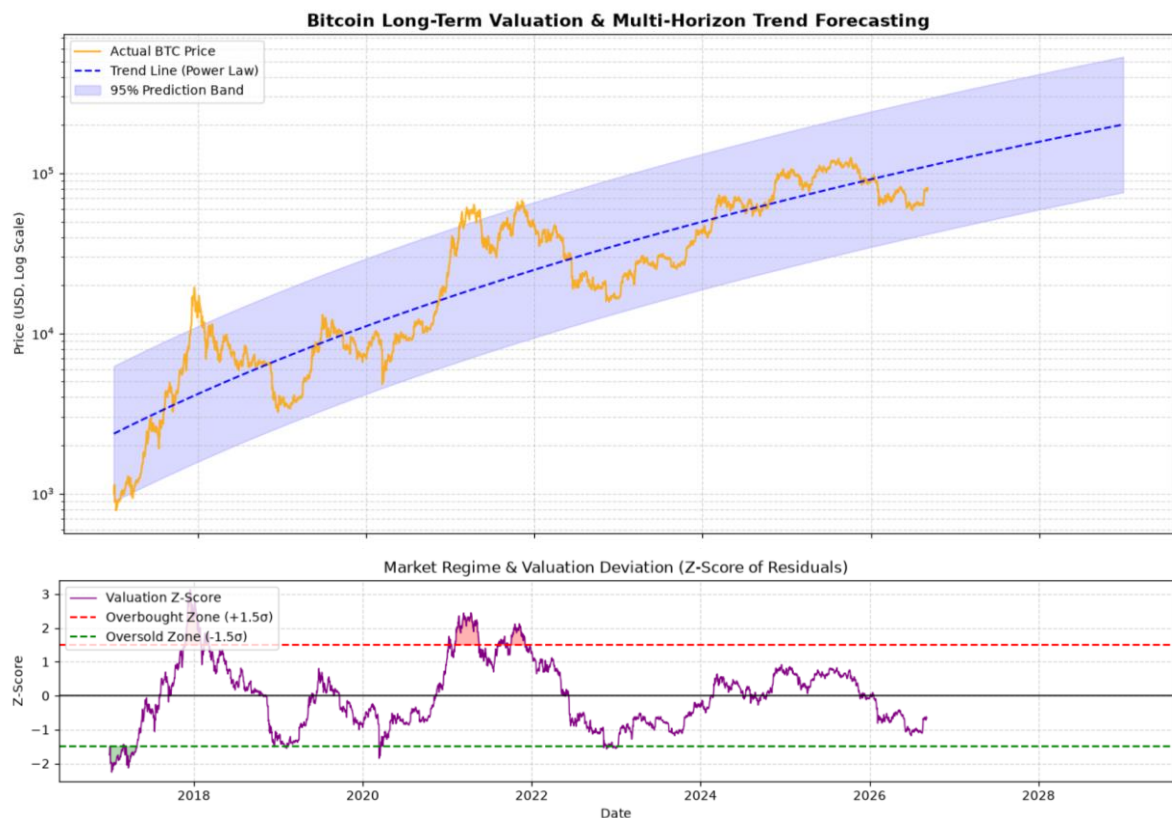


Figure 2. Long-term trend with residual Z-score valuation diagnostic.

Actual price: $\approx \$79,700$

Model-implied trend value: $\approx \$110,700$

Valuation Z-score: -0.66

Interpretation: Moderately below the long-term trend, still within a normal range (neither extreme overbought nor oversold).

Stage 3 – Hybrid Model

Table 3. Hybrid Short-term Model Performance

Metric	Value
R-squared	0.005
Adjusted R-squared	0.003
Number of Observations	3,505
F-statistic	3.45
Prob (F-statistic)	0.004

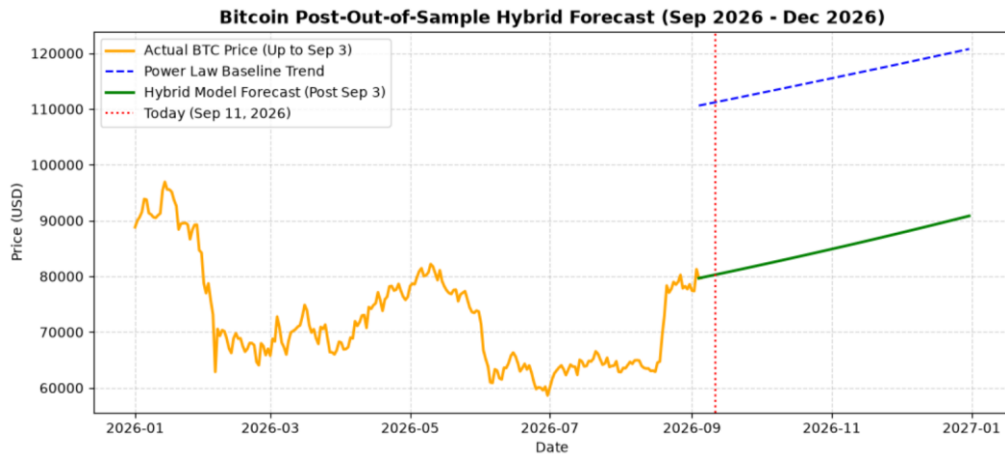


Figure 3. Hybrid short-term forecast versus pure power-law trend (Sep–Dec 2026).

Table 4. Hybrid Forecast Examples

Date	Power-Law Baseline	Hybrid Forecast	Projected Z-Score
11 Sep 2026	\$111,196	\$80,267	−0.66
30 Sep 2026	\$112,791	\$81,943	−0.64
31 Oct 2026	\$115,430	\$84,788	−0.62

The short-term residual model achieved an R-squared of only 0.005.

Most short-term factors were statistically weak.

As a result, the hybrid forecast stayed close to a slow mean-reversion path from the current price toward the long-term trend.

5. Interpretation & Insights

The power-law model provides a useful long-term anchor. An R-squared of 0.83 is relatively high for financial price data and suggests that a meaningful portion of Bitcoin’s historical price path can be described by time since inception.

The residual Z-score adds practical value by translating model error into a simple “expensive / cheap” reading. Historically, readings above +1.5 often coincided with cycle tops, while readings near or below −1.5 appeared near major bottoms.

The attempt to improve short-term forecasts with volume and return factors was largely unsuccessful. This is an important finding: Bitcoin’s short-term movements contain a high degree of noise that simple linear factors struggle to capture. In this case, adding complexity did not improve explanatory power.

Best-performing component: The original power-law model ($R^2 = 0.83$).

The later stages improved interpretability and diagnostic ability rather than raw predictive accuracy.

6. Limitations

The power-law relationship is an empirical observation, not a theoretical law. It may weaken or change in the future.

Residuals show strong autocorrelation, so standard OLS inference should be treated with caution.

The short-term hybrid model has almost no explanatory power and should not be used for precise near-term predictions.

The framework is intentionally simple. It does not incorporate on-chain metrics, macroeconomic variables, or regime-switching dynamics.

7. Real-World Relevance

For investment or research teams, this type of work offers three practical contributions:

A transparent long-term reference point – useful for framing multi-year scenarios.

A quantitative valuation context – helps avoid purely narrative-driven decisions about whether the asset is “cheap” or “expensive.”

A clear demonstration of model limits – showing where statistical tools add value and where they stop being useful.

In Web3 data or crypto research roles, the ability to build, test, and honestly evaluate such frameworks is often more valuable than producing a single high- R^2 model that cannot be explained.

8. What This Project Demonstrates

Ability to move from a research question to a structured analytical pipeline

Practical application of regression, residual analysis, and standardization

Willingness to iterate and critically assess whether added complexity improves results

Clear communication of both strengths and weaknesses of the approach

Although my coding background is not extensive, the project shows that solid statistical reasoning and structured problem-solving can still produce useful, interpretable analysis.

9. Conclusion

This project confirms that Bitcoin exhibits a relatively strong long-term power-law trend ($R^2 \approx 0.83$). The residual Z-score provides a practical way to gauge current valuation relative to that trend. Attempts to refine the model with short-term market factors added little predictive power, reinforcing that long-term structure is more reliable than short-term noise in this dataset.

The final framework is simple, transparent, and honest about its limitations — qualities that matter in applied research and investment analysis.

Tools used: Python (pandas, statsmodels, matplotlib), statistical concepts from applied regression and residual analysis.

Feel free to reach out if you would like to discuss the methodology or results in more detail.